FORCE AND LAWS OF MOTION

Class 9 – Easy, exam-ready answers based on the questions in the uploaded pages

Note: Repeated questions from the different sections have been combined. Only the unique concepts/questions are included, so the same answer is not repeated unnecessarily.

A. VERY SHORT ANSWER / ONE-TO-TWO LINE QUESTIONS

1. What is force?

Answer: Force is a push or a pull that can change the state of rest or motion of an object, or change its shape.

2. What are the main effects of a force?

Answer: A force can start or stop motion, change the speed or direction of motion, or change the shape of an object.

3. What is momentum? Give its SI unit.

Answer: Momentum is the quantity of motion possessed by a body. Momentum = mass x velocity. Its SI unit is kg m/s.

4. What is the momentum of a body at rest?

Answer: It is zero because its velocity is zero. p = mv = m x 0 = 0.

5. Is momentum a scalar or vector quantity?

Answer: Momentum is a vector quantity because it has both magnitude and direction.

6. Which physical quantity measures the inertia of a body?

Answer: Mass measures the inertia of a body. Greater mass means greater inertia.

7. What is inertia?

Answer: Inertia is the tendency of a body to resist any change in its state of rest or motion, or in its direction of motion.

8. Name the three types of inertia.

Answer: The three types are: inertia of rest, inertia of motion, and inertia of direction.

9. What is the other name of Newton's first law of motion?

Answer: Newton's first law is also called the law of inertia.

10. What is a balanced force?

Answer: Balanced forces are equal and opposite forces whose resultant force is zero. They do not produce acceleration.

11. What is an unbalanced force?

Answer: Unbalanced forces have a non-zero resultant force and can change the motion of a body.

12. What force is required to maintain uniform motion in a straight line?

Answer: Zero net force is required. If the body is already moving uniformly, no unbalanced force is needed.

13. State Newton's second law of motion.

Answer: The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force.

14. What physical quantity corresponds to the rate of change of momentum?

Answer: Force corresponds to the rate of change of momentum.

15. Define one newton of force.

Answer: One newton is the force that produces an acceleration of 1 m/s^2 in a body of mass 1 kg.

16. State Newton's third law of motion.

Answer: For every action, there is an equal and opposite reaction. The two forces act on different bodies.

17. How are action and reaction forces related?

Answer: They are equal in magnitude and opposite in direction, and they act on two different bodies.

18. What is recoil velocity?

Answer: Recoil velocity is the velocity with which a gun moves backward when a bullet is fired forward.

19. Which law explains swimming?

Answer: Swimming is explained by Newton's third law of motion.

20. What happens to momentum if velocity is doubled, keeping mass constant?

Answer: Momentum is doubled because $p = mv$ and mass remains constant.

21. What happens to acceleration if mass is doubled for the same force?

Answer: Acceleration becomes half because $F = ma$, so $a = F/m$.

22. What happens to the momentum if both mass and velocity are doubled?

Answer: Momentum becomes four times the original value. Therefore, the percentage increase is 300%.

23. Why does the speed of a dropped ball increase as it falls?

Answer: Earth's gravitational force acts downward on the ball and gives it acceleration, so its speed increases.

24. What is the momentum of a ball at the highest point when thrown vertically upward?

Answer: It is zero because the velocity of the ball is zero at the highest point.

25. Why is the jerk smaller when a gun is heavy?

Answer: For the same bullet momentum, a heavier gun gets a smaller backward velocity and hence a smaller recoil/jerk.

26. Why does a moving box need friction equal to the applied force when it moves at constant velocity?

Answer: At constant velocity, acceleration is zero, so the net force is zero. Therefore, friction balances the applied force: friction = F in the opposite direction.

B. SHORT ANSWER QUESTIONS

1. Why do passengers fall backward when a bus starts suddenly?

Answer: When the bus starts suddenly, the feet of the passengers move with the bus but their upper bodies tend to remain at rest due to inertia of rest. Hence they fall backward.

2. Why does a person fall forward when a moving bus stops suddenly?

Answer: The feet stop with the bus, but the upper body tends to continue moving forward due to inertia of motion. Hence the person falls forward.

3. Why does a passenger move sideways when a bus takes a sharp turn?

Answer: The bus changes its direction, but the passenger's body tends to continue in the original straight direction due to inertia of direction. Therefore, the passenger leans or moves sideways.

4. Why is it easier to walk on a rough surface than on a slippery surface?

Answer: A rough surface provides more friction. Friction gives our feet a firm grip on the ground, allowing us to push the ground backward and move forward. A slippery surface provides less friction.

5. Explain the law of inertia and give examples.

Answer: The law of inertia states that a body continues in its state of rest or uniform straight-line motion unless an unbalanced external force acts on it. Examples: passengers fall backward when a bus starts suddenly; dust comes out when a carpet is beaten.

6. Why is it difficult to hold a hose-pipe that throws water at high speed?

Answer: Water is thrown forward with high momentum. By Newton's third law, the water produces an equal and opposite reaction on the hose, causing it to push backward strongly.

7. Why is it easier to stop a tennis ball than a cricket ball moving at the same speed?

Answer: At the same speed, the cricket ball has greater mass and therefore greater momentum. A larger momentum requires a larger change in momentum to stop, so the cricket ball is harder to stop.

8. Why is it dangerous to jump out of a moving bus?

Answer: After jumping out, the feet may stop or slow down when they touch the ground, but the upper body tends to continue moving forward due to inertia of motion. This can cause a fall and injury.

9. Why are road accidents at high speeds generally more severe than at low speeds?

Answer: At higher speed, a vehicle has greater momentum and kinetic energy. During a collision, a larger change in momentum can occur in a short time, producing a larger force and generally more severe effects.

10. Why does a gun give a backward jerk when a bullet is fired?

Answer: Before firing, the total momentum is zero. The bullet moves forward after firing, so the gun moves backward to conserve total momentum. This backward motion is recoil.

11. Why does a boat move backward when a person jumps from it to the shore?

Answer: The person pushes the boat backward while jumping forward. The boat receives an equal and opposite reaction, so it moves backward.

12. Why does a boatman push the bank with an oar to move the boat away from the bank?

Answer: The boatman pushes the bank backward with the oar. The bank exerts an equal and opposite force on the oar, pushing the boat away from the bank.

13. Why can a cart move when a horse pulls it even though action and reaction are equal?

Answer: The horse pushes the ground backward with its feet. The ground exerts a forward frictional force on the horse. This external force moves the horse and the cart forward. The action-reaction pair acts on different bodies.

14. Why does the Earth have very small acceleration when a ball falls toward it?

Answer: The ball and Earth exert equal and opposite gravitational forces on each other. Because Earth's mass is extremely large, the same force produces a very small acceleration of Earth.

15. Why do fruits or leaves fall when a tree branch is shaken suddenly?

Answer: The branch moves suddenly, but the fruits or leaves tend to remain in their original state due to inertia of rest. As a result, they may detach and fall.

16. Why does dust come out when a carpet is beaten with a stick?

Answer: When the carpet is beaten, it moves suddenly but the dust particles tend to remain at rest due to inertia. They separate from the carpet and fall away.

17. Why does an athlete run some distance before taking a long jump?

Answer: Running increases the athlete's speed. Due to greater speed and momentum, the athlete can cover a greater horizontal distance after jumping.

18. Why does a person get more hurt when falling on a cemented road than on a soft surface?

Answer: A hard surface stops the body in a very short time. For the same change in momentum, a shorter stopping time produces a larger force. A soft surface increases the stopping time and reduces the force.

19. Why does a cricket player move his hands backward while catching a fast ball?

Answer: Moving the hands backward increases the time taken to stop the ball. Since force depends on the rate of change of momentum, increasing stopping time reduces the force on the hands.

20. Why does a small car accelerate more easily than a large car for the same force?

Answer: A small car usually has less mass. From a = F/m, for the same force, smaller mass gives greater acceleration.

21. On what factor does the inertia of a body depend? Which has more inertia: a cricket ball or a rubber ball of the same size?

Answer: Inertia depends on mass. A cricket ball has greater mass than a rubber ball of the same size, so it has greater inertia.

C. NUMERICAL / FORMULA-BASED QUESTIONS

1. Calculate the momentum of a toy car of mass 300 g moving with a speed of 18 km/h.

Answer: Given: m = 300 g = 0.3 kg; v = 18 km/h = 5 m/s.
Momentum, p = mv = 0.3 x 5 = **1.5 kg m/s**.
Answer: 1.5 kg m/s.

2. A body of mass 30 kg has a momentum of 150 kg m/s. What is its velocity?

Answer: Using p = mv, v = p/m = 150/30 = **5 m/s**.
Answer: 5 m/s.

3. A force of 100 N acts on a mass of 25 kg for 5 s. What velocity does it generate if the body starts from rest?

Answer: Using F = ma: a = F/m = 100/25 = 4 m/s^2.
Using v = u + at, with u = 0: v = 0 + (4 x 5) = **20 m/s**.
Answer: 20 m/s.

4. A body of mass 5 kg is moving with a velocity of 10 m/s. A force acts for 25 s and the velocity becomes 35 m/s. Find the force.

Answer: Given: m = 5 kg, u = 10 m/s, v = 35 m/s, t = 25 s.
a = (v - u)/t = (35 - 10)/25 = 1 m/s^2.
F = ma = 5 x 1 = **5 N**.
Answer: 5 N.

5. Show that kg m/s and N s represent the same quantity.

Answer: 1 N = 1 kg m/s^2.
Therefore, 1 N s = (1 kg m/s^2) x s = **1 kg m/s**.
Thus, kg m/s and N s are equivalent units of **linear momentum**.

Important formulas for Class 9
Momentum: p = m x v
Force: F = m x a
Acceleration: a = (v - u) / t
Newton's second law: F = change in momentum / time
1 N = 1 kg m/s^2
1 N s = 1 kg m/s

D. LONG ANSWER QUESTIONS

1. Explain balanced and unbalanced forces with examples.

Answer: When two or more forces act on a body, their combined effect is called the resultant force.

Balanced forces: If the resultant force is zero, the forces are balanced. They do not produce acceleration. For example, a book lying on a table has its weight downward and the table's support force upward. These forces balance each other.

Unbalanced forces: If the resultant force is not zero, the forces are unbalanced. They can change the speed or direction of motion. For example, when a football is kicked, the applied force changes its state of motion.

2. Explain friction and its role in the motion of objects.

Answer: Friction is a force that opposes the relative motion or tendency of motion between two surfaces in contact.

Useful effects: It helps us walk, lets vehicles move without slipping, allows us to write, and helps brakes stop vehicles.
Harmful effects: It causes wear and tear of machine parts and produces unwanted heat.

Thus, friction is sometimes necessary and sometimes undesirable. Rough surfaces generally provide more friction than smooth surfaces.

3. State Newton's three laws of motion.

Answer: First law: A body remains at rest or continues in uniform straight-line motion unless acted upon by an unbalanced external force. It is also called the law of inertia.

Second law: The rate of change of momentum of a body is directly proportional to the applied force and occurs in the direction of the force. For constant mass, F = ma.

Third law: For every action, there is an equal and opposite reaction. The action and reaction act on different bodies.

4. State Newton's second law and obtain the expression F = ma.

Answer: Suppose a body of mass m has initial velocity u and final velocity v in time t.

Initial momentum = mu
Final momentum = mv
Change in momentum = mv - mu = m(v-u)
Rate of change of momentum = m(v-u)/t

But acceleration, a = (v-u)/t.
Therefore, rate of change of momentum = ma.
According to Newton's second law, force is proportional to the rate of change of momentum. In SI units, the proportionality constant is 1.
Therefore, F = ma.

5. Explain Newton's third law and describe a simple experimental demonstration.

Answer: Newton's third law states: **For every action, there is an equal and opposite reaction.** The two forces act on different bodies.

Experiment: Take two spring balances and hook them together. Pull one balance while the other is held or pulled in the opposite direction. Both balances show equal readings (for the same interaction). This demonstrates that the forces on the two balances are equal in magnitude and opposite in direction.

Other examples include walking, swimming, rocket motion and gun recoil.

6. Give three applications of Newton's third law of motion.

Answer: 1. **Walking:** We push the ground backward; the ground pushes us forward.
2. **Swimming:** A swimmer pushes water backward; water pushes the swimmer forward.
3. **Rocket motion:** Hot gases are pushed downward/backward; the reaction force pushes the rocket upward/forward.

7. Define momentum and state the factors on which it depends.

Answer: Momentum is the quantity of motion of a body. It is given by **p = mv**, where m is mass and v is velocity.

Therefore, momentum depends on two factors: **mass** and **velocity**. Greater mass or greater velocity gives greater momentum when the other factor is kept constant. Its SI unit is **kg m/s**, and it is a vector quantity.

8. Explain why road accidents at high speed can be more severe than at low speed.

Answer: A moving vehicle has momentum p = mv. At higher speed, its momentum is greater. Also, its kinetic energy increases with the square of speed. During a collision, a large change in momentum may occur in a short time, producing a large force. Therefore, accidents at high speeds can cause more severe damage and injuries.

9. Explain why a cricket player moves his hands backward while catching a fast cricket ball.

Answer: The ball has momentum and must be brought to rest. The player moves the hands backward to increase the time taken to stop the ball. From Newton's second law, force = change in momentum / time. For the same change in momentum, increasing the stopping time decreases the force on the hands. Thus, moving the hands backward helps reduce the chance of injury.

10. Show how Newton's first law can be obtained from Newton's second law.

Answer: Newton's second law gives F = ma. If no external unbalanced force acts on a body, F = 0.
So, ma = 0. For a body of non-zero mass, a = 0.
Zero acceleration means the velocity remains constant. Therefore, a body at rest remains at rest, and a moving body continues with uniform velocity in a straight line unless an external unbalanced force acts on it.

This is Newton's first law of motion.

Exam tip: For Class 9 answers, write the law/definition first, then give the reason or formula, and finally write the conclusion. For numerical questions, always write Given → Formula → Substitution → Answer with unit.